AI-Assisted Coding

Assignment-3.5

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Batch:45

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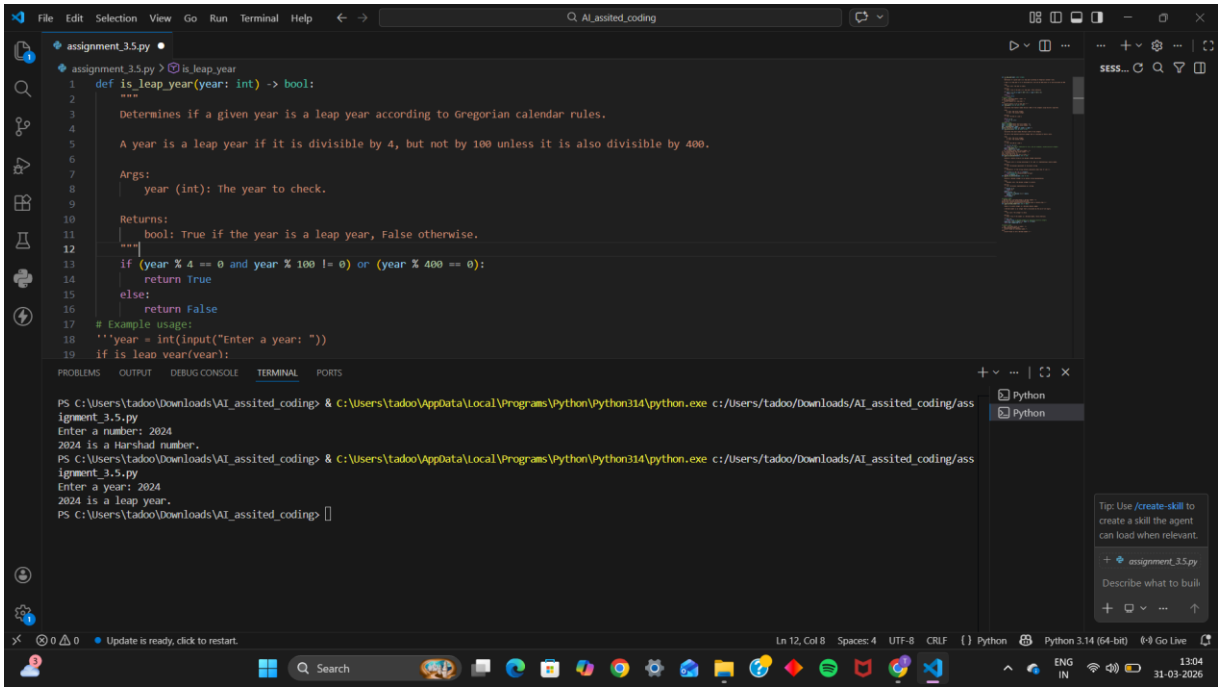
Question 1: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year.

Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions.



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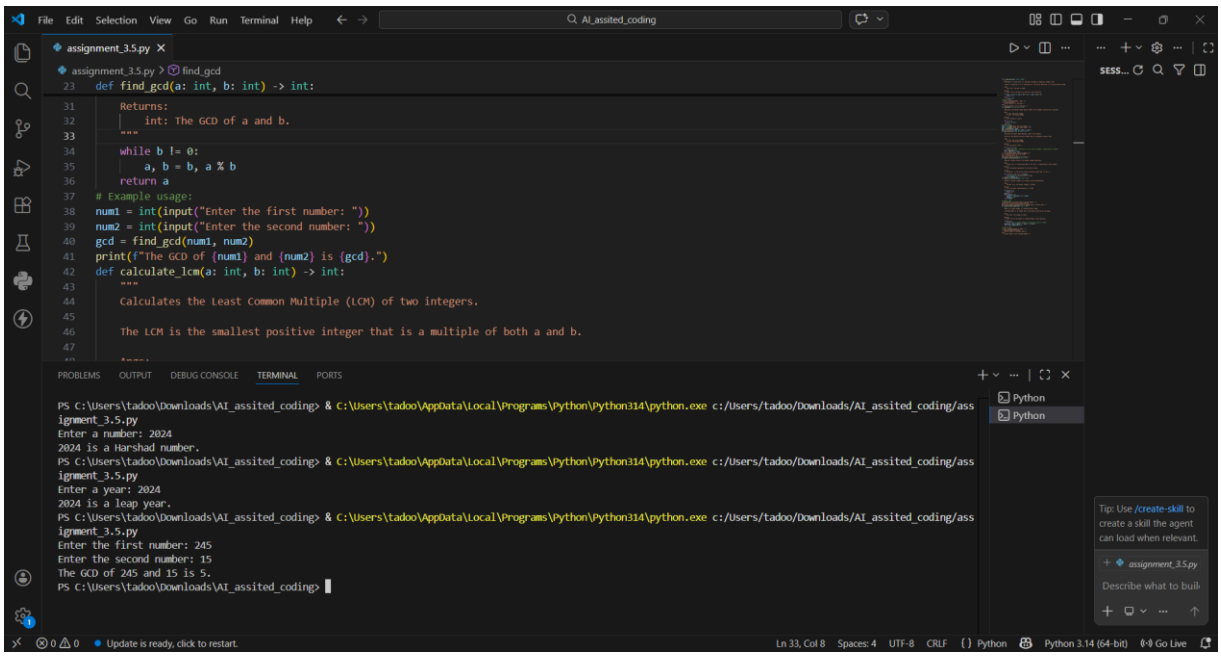
Question 2: One Shot Prompting (GCD of Two Numbers) Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6 Task:

• Compare with a zero-shot solution.

• Analyze algorithm efficiency



Question 3: Few Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

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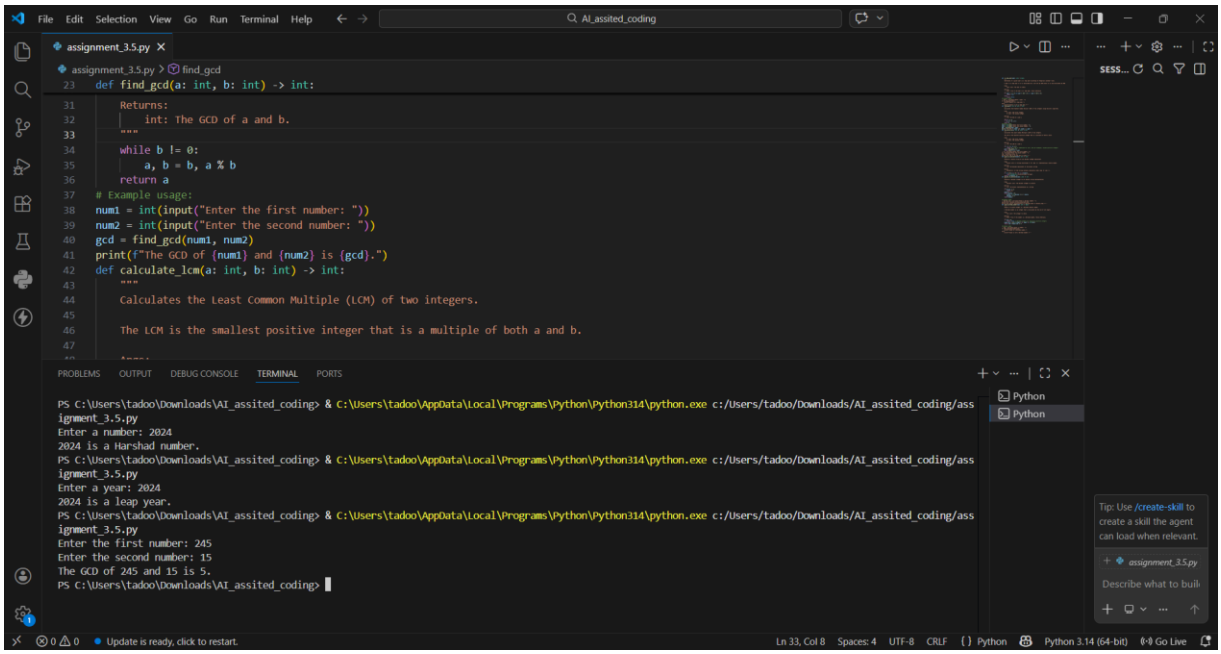
• Input: 4, 6 → Output: 12

• Input: 5, 10 → Output: 10

• Input: 7, 3 → Output: 21 Task:

• Examine how examples guide formula selection.

• Test edge cases



Question 5: One Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010 Task:

• Compare clarity with zero-shot output.

• Analyze handling of zero and negative numbers.

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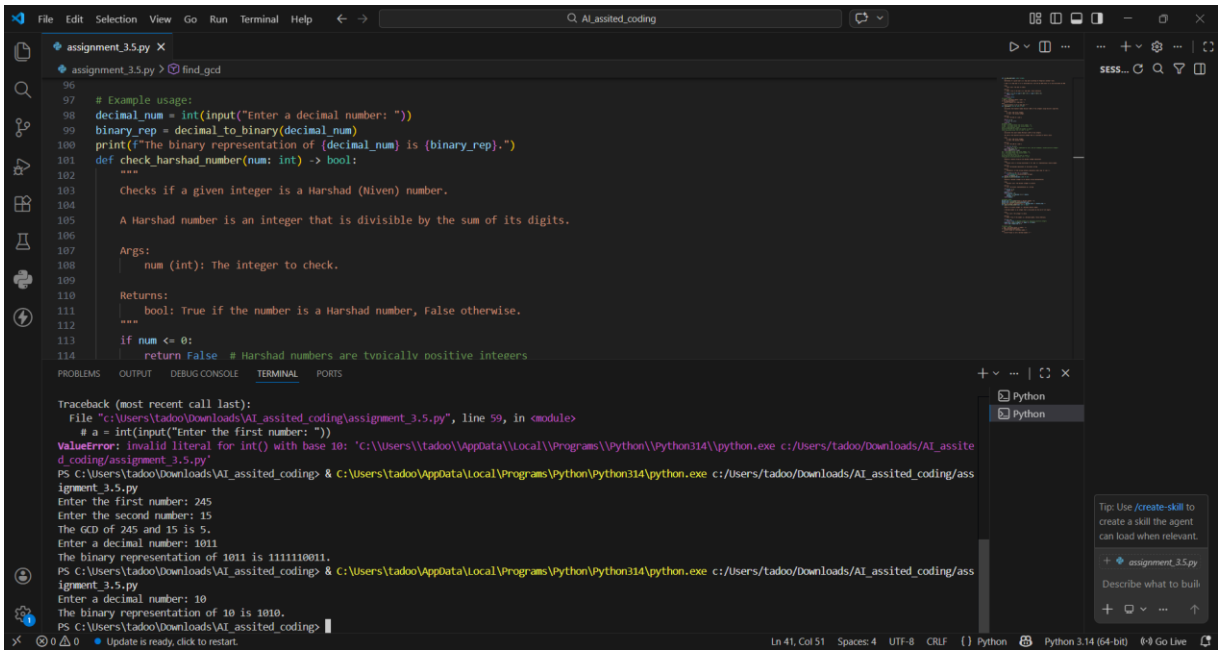

Question 6: Few Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks whether a number is a Harshad (Niven) number.

Examples:

- Input: 18 → Output: Harshad Number
- Input: 21 → Output: Harshad Number • Input: 19 → Output: Not a Harshad Number Task:
- Test boundary conditions.
- Evaluate robustness

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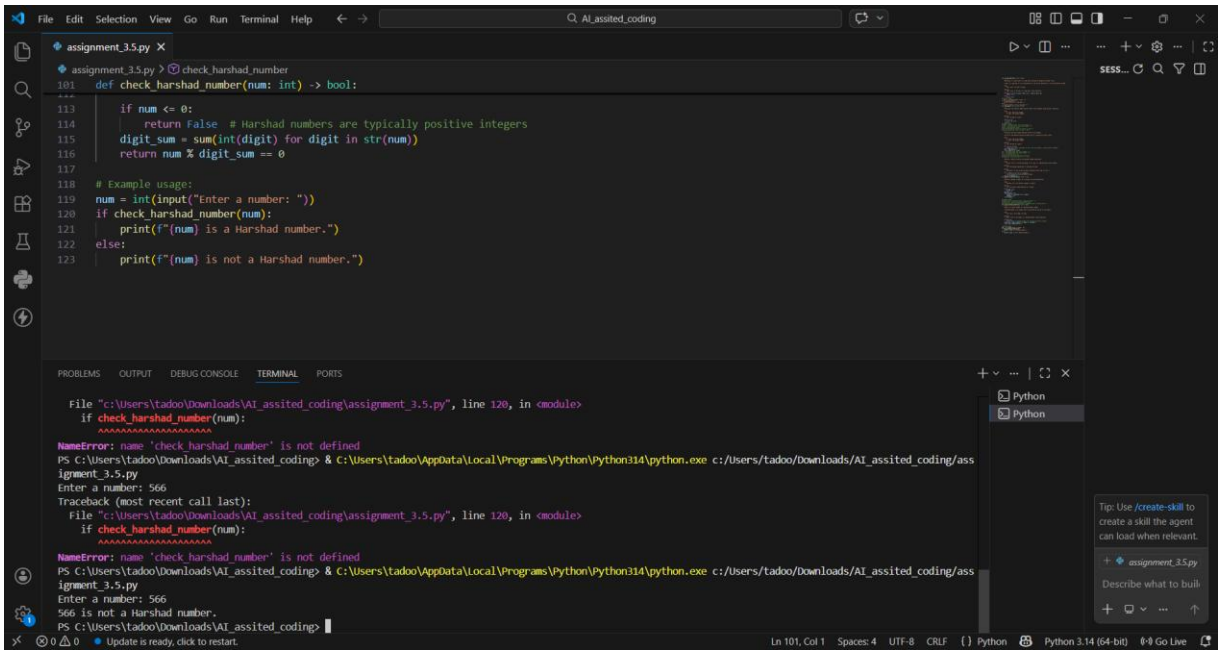
File Edit Selection View Go Run Terminal Help
AI_assited_coding
assignment_3.5.py
assignment_3.5.py > check_harshad_number
def check_harshad_number(num: int) -> bool:
if num <= 0:
return False # Harshad numbers are typically positive integers
digit_sum = sum(int(digit) for digit in str(num))
return num % digit_sum == 0
# Example usage:
num = int(input("Enter a number: "))
if check_harshad_number(num):
print(f"{num} is a Harshad number.")
else:
print(f"{num} is not a Harshad number.")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
File "c:\Users\tadoo\Downloads\AI_assited_coding\assignment_3.5.py", line 120, in <module>
if check_harshad_number(num):
NameError: name 'check_harshad_number' is not defined
PS C:\Users\tadoo\Downloads\AI_assited_coding> & C:\Users\tadoo\AppData\Local\Programs\Python\Python314\python.exe c:/Users/tadoo/Downloads/AI_assited_coding/assignment_3.5.py
Enter a number: 566
Traceback (most recent call last):
File "c:\Users\tadoo\Downloads\AI_assited_coding\assignment_3.5.py", line 120, in <module>
if check_harshad_number(num):
NameError: name 'check_harshad_number' is not defined
PS C:\Users\tadoo\Downloads\AI_assited_coding> & C:\Users\tadoo\AppData\Local\Programs\Python\Python314\python.exe c:/Users/tadoo/Downloads/AI_assited_coding/assignment_3.5.py
Enter a number: 566
566 is not a Harshad number.
PS C:\Users\tadoo\Downloads\AI_assited_coding>
Python
Python
Tip: Use /create-skill to create a skill the agent can load when relevant.
Describe what to build
Update is ready, click to restart.
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